I'll create the **Study / Paper Forensics Expansion Pack v0.2** with 7 carefully curated cases covering the requested distribution and domain spread.

---

1. Executive Summary

This expansion pack adds **7 high-signal forensic cases** designed to address specific weaknesses identified in the v0.1 pilot:

What this pack adds:
- **2 clear overclaim cases** with unambiguous rhetorical inflation (Technology AI productivity claims; Workplace RTO mandate evidence)
- **2 solid cases** demonstrating proper causal caution and uncertainty quantification (Education RCT; Environmental monitoring validation)
- **1 mechanism vs. evidence case** highlighting the gap between biological plausibility and clinical efficacy (Behavioral neuroscience)
- **1 correlation vs. causation case** showing classic econometric overreach (Labor economics)
- **1 borderline case** where evidence is genuinely mixed and reasonable analysts could disagree (Public policy UBI pilots)

Domain expansion: Shifted from v0.1's heavy health/supplement focus to include education, environmental science, technology/AI, workplace policy, economics, and public policy while maintaining accessibility.

Quality improvements: All cases feature clean citation-free prose, explicit gold justifications, and diagnostic evidence spans. Overclaim cases are sharper and less debatable than the pilot's vitamin D example.

---

2. Dataset Table

case_id	title	bucket	domain	source_type	gold_case_type	expected_main_failure	difficulty
v0.2_001	Phonics Intervention RCT	solid	education	synthetic_summary	clear	none	easy
v0.2_002	Low-Cost Air Sensor Validation	solid	environmental_science	synthetic_summary	clear	none	easy
v0.2_003	AI Coding Assistant Productivity	overclaim	technology/AI	synthetic_press_summary	clear	survivorship_bias + generalization	medium
v0.2_004	Return-to-Office Collaboration	overclaim	workplace_policy	synthetic_press_summary	clear	selection_bias + reverse_causation	medium
v0.2_005	Dopamine Habit Formation	mechanism_vs_evidence	behavioral_neuroscience	synthetic_summary	clear	mechanistic_inflation	medium

| v0.2_006 | Minimum Wage Automation | correlation_vs_causation | economics | synthetic_press_summary | clear | causal_overclaim | easy |
| v0.2_007 | Universal Basic Income Pilots | mixed | public_policy | synthetic_summary | borderline | external_validity | hard |

---

3. Full Case Definitions

Case v0.2_001: Phonics Intervention RCT

Metadata:
```yaml
case_id: v0.2_001
title: "Phonics Intervention RCT"
domain: education
category_bucket: solid
difficulty: easy
source_type: synthetic_summary
expected_strength: strong
expected_main_failure: none
```

Input:
```yaml
input_text: |
  A randomized controlled trial evaluated a systematic phonics intervention for first-grade students struggling with reading. Researchers randomly assigned 240 students across 12 schools to either the phonics program (n=120) or business-as-usual instruction (n=120). The intervention involved 30 minutes of daily structured phonics instruction delivered by trained teaching assistants over 12 weeks.

  Students in the intervention group showed significant improvement on standardized word reading tests (Cohen's d = 0.42, 95% CI: 0.18 to 0.66, p = 0.001). The effect was larger for students with baseline reading scores below the 25th percentile (d = 0.61) compared to those between the 25th and 40th percentiles (d = 0.28). No significant effects were observed for reading comprehension measures (d = 0.15, 95% CI: -0.09 to 0.39, p = 0.22).

  The authors note that effects were assessed immediately post-intervention and durability is unknown. They also acknowledge that implementation fidelity varied across schools (80-95% of sessions completed as intended) and that results may not generalize to schools with fewer resources for teaching assistant support.
```

Gold Layer:
```yaml
gold_label: solid
gold_confidence: high
```

gold_case_type: clear
gold_summary: |
  Well-conducted RCT with appropriate effect size reporting, confidence intervals, subgroup analysis with proper interaction testing, and explicit limitations regarding durability and generalizability.
gold_justification: |
  The study demonstrates proper causal inference methodology (randomization, control group, pre-registered analysis). Effect sizes are modest and interpreted appropriately. Authors explicitly acknowledge boundary conditions (immediate post-test only, implementation variance, resource requirements) without defensive minimization. The differential effect by baseline ability is properly analyzed with interaction terms rather than cherry-picked subgroups.
gold_evidence_span: "effects were assessed immediately post-intervention and durability is unknown"
gold_alternative_reading: null
```

**Expected Findings:**
```yaml
expected_findings:
  - finding: appropriate_uncertainty_quantification
    severity: none
    confidence: high
  - finding: proper_subgroup_analysis
    severity: none
    confidence: high
  - finding: honest_limitations
    severity: none
    confidence: high
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "95% CI: 0.18 to 0.66"
    why_it_matters: Reports uncertainty interval rather than just point estimate
    supports_finding: proper_uncertainty_quantification
  - span: "effects were assessed immediately post-intervention and durability is unknown"
    why_it_matters: Explicit boundary condition on temporal generalizability
    supports_finding: honest_limitations
  - span: "implementation fidelity varied across schools (80-95% of sessions completed as intended)"
    why_it_matters: Acknowledges real-world implementation variance without dismissing it
    supports_finding: honest_limitations
```

Diagnostic Value:
```yaml
```
```

why_this_case_is_useful: |
  Baseline example of proper educational RCT reporting. Tests whether systems can recognize appropriate caution when it is present, not just flag overclaims when absent. The explicit limitation statements are models of honest scientific communication.
```

---

### Case v0.2_002: Low-Cost Air Sensor Validation

**Metadata:**
```yaml
case_id: v0.2_002
title: "Low-Cost Air Sensor Validation"
domain: environmental_science
category_bucket: solid
difficulty: easy
source_type: synthetic_summary
expected_strength: strong
expected_main_failure: none
```

Input:
```yaml
input_text: |
  Researchers validated low-cost particulate matter sensors against regulatory-grade reference instruments at three urban monitoring sites over six months. The study compared readings from 45 low-cost sensors (costing approximately $250 each) against Federal Reference Method monitors for PM2.5 concentrations ranging from 5 to 85 micrograms per cubic meter.

  Linear regression showed strong correlation between low-cost and reference sensors ($R^2$ = 0.87, slope = 0.94, intercept = 1.2 μg/m³). However, accuracy degraded significantly at humidity levels above 80%, with measurement error increasing from ±3 μg/m³ to ±8 μg/m³. Sensors also underestimated concentrations during wildfire smoke events (PM2.5 > 60 μg/m³) by an average of 15%.

  The authors recommend these sensors for identifying spatial patterns and temporal trends in typical urban conditions but caution against using them for regulatory compliance monitoring or health studies requiring precise exposure assessment during extreme pollution events. They provide site-specific calibration equations and note that sensors require recalibration every 12 months.
```

Gold Layer:
```yaml
gold_label: solid
gold_confidence: high
```
```

gold_case_type: clear
gold_summary: |
  Rigorous validation study with explicit boundary conditions, degradation analysis under stress conditions, and appropriate usage recommendations that respect instrument limitations.
gold_justification: |
  The study demonstrates excellent metrological practice: correlation is not treated as interchangeability, specific failure modes are characterized (humidity, high concentration), and recommendations are appropriately bounded. The R² of 0.87 is strong but not overstated as perfect agreement. Authors explicitly forbid inappropriate use cases (regulatory compliance) despite potential demand for such applications.
gold_evidence_span: "recommend these sensors for identifying spatial patterns and temporal trends... but caution against using them for regulatory compliance"
gold_alternative_reading: null
```

**Expected Findings:**
```yaml
expected_findings:
  - finding: appropriate_technical_validation
    severity: none
    confidence: high
  - finding: explicit_boundary_conditions
    severity: none
    confidence: high
  - finding: usage_guidance_respects_limitations
    severity: none
    confidence: high
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "accuracy degraded significantly at humidity levels above 80%"
    why_it_matters: Identifies specific environmental conditions where instrument fails
    supports_finding: explicit_boundary_conditions
  - span: "underestimated concentrations during wildfire smoke events... by an average of 15%"
    why_it_matters: Critical failure mode during high-stakes measurement scenarios
    supports_finding: appropriate_technical_validation
  - span: "caution against using them for regulatory compliance monitoring"
    why_it_matters: Explicit prohibition of inappropriate use despite potential market pressure
    supports_finding: usage_guidance_respects_limitations
```

Diagnostic Value:
```yaml
why_this_case_is_useful: |
```
```

  Tests recognition of proper instrumental validation and appropriate epistemic boundaries. The explicit "do not use for X" recommendation is a key signal of scientific integrity that forensic systems should identify and reward.
```

---

### Case v0.2_003: AI Coding Assistant Productivity

**Metadata:**
```yaml
case_id: v0.2_003
title: "AI Coding Assistant Productivity"
domain: technology/AI
category_bucket: overclaim
difficulty: medium
source_type: synthetic_press_summary
expected_strength: weak
expected_main_failure: survivorship_bias_generalization
```

Input:
```yaml
input_text: |
  A comprehensive study of AI-powered coding assistants reveals transformative productivity gains for software development teams. The research tracked 450 developers using an AI code completion tool over three months, measuring code commits, pull request completion times, and feature delivery velocity.

  Developers using the AI assistant showed 42% faster completion times on coding tasks and produced 35% more code commits per week compared to baseline measurements taken before tool adoption. The study concludes that AI coding assistants fundamentally enhance developer productivity and will reshape software engineering workflows across the industry. The researchers recommend immediate enterprise adoption to maintain competitive advantage.

  Notably, 78% of developers initially enrolled in the study continued active use of the tool through the study period, indicating high satisfaction and sustained utility. The productivity gains were consistent across both junior and senior developers, suggesting universal applicability regardless of experience level.
```

Gold Layer:
```yaml
gold_label: overclaim
gold_confidence: high
gold_case_type: clear
gold_summary: |
```
```

Study suffers from severe survivorship bias (22% attrition likely non-random), lack of control group, regression to the mean in self-selected early adopters, and inappropriate generalization from voluntary adoption to mandatory enterprise deployment.
gold_justification: |
The 22% attrition rate is mentioned but dismissed as "high satisfaction" rather than acknowledged as potential selection bias. No control group exists—productivity gains could reflect Hawthorne effects, seasonal variation, or project complexity differences. "Continued active use" filters out users who tried the tool and abandoned it due to negative experiences. The jump from "voluntary adopters in study" to "enterprise adoption recommendations" ignores the difference between self-selected tool enthusiasts and mandated users. The claim that gains were "consistent across junior and senior developers" lacks interaction testing and raw data.
gold_evidence_span: "78% of developers initially enrolled in the study continued active use"
gold_alternative_reading: |
The 22% who discontinued may represent users who found the tool counterproductive or frustrating. Early voluntary adopters likely differ systematically from reluctant enterprise-mandated users. Without a control group, the 42% improvement could be entirely artifactual.
```

**Expected Findings:**
```yaml
expected_findings:
  - finding: survivorship_bias
    severity: high
    confidence: high
  - finding: lack_of_control_group
    severity: high
    confidence: high
  - finding: inappropriate_generalization
    severity: high
    confidence: high
  - finding: self_selection_bias
    severity: medium
    confidence: high
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "78% of developers initially enrolled in the study continued active use"
    why_it_matters: 22% attrition likely non-random (frustrated users quit), creating survivorship bias
    supports_finding: survivorship_bias
  - span: "compared to baseline measurements taken before tool adoption"
    why_it_matters: No concurrent control group; cannot distinguish tool effect from Hawthorne/seasonality
    supports_finding: lack_of_control_group
```
```

  - span: "recommend immediate enterprise adoption"
    why_it_matters: Generalizes from voluntary early adopters to mandatory enterprise users without justification
    supports_finding: inappropriate_generalization
```

**Diagnostic Value:**
```yaml
why_this_case_is_useful: |
  Classic example of technology hype cycle research. Tests ability to spot survivorship bias framed as success metrics and inappropriate causal leaps from observational adoption data to prescriptive enterprise policy. The rhetorical framing ("transformative," "fundamentally enhance," "immediate adoption") signals marketing language masquerading as science.
```

Case v0.2_004: Return-to-Office Collaboration

Metadata:
```yaml
case_id: v0.2_004
title: "Return-to-Office Collaboration"
domain: workplace_policy
category_bucket: overclaim
difficulty: medium
source_type: synthetic_press_summary
expected_strength: weak
expected_main_failure: selection_bias_reverse_causation
```

Input:
```yaml
input_text: |
  New research demonstrates that return-to-office mandates significantly improve workplace collaboration and innovation. Analyzing data from 32 companies that implemented mandatory in-office policies between 2022 and 2024, the study found that employees in office-mandated firms sent 23% more emails to colleagues outside their immediate teams and participated in 31% more cross-functional meetings compared to the previous remote-work period.

  Companies with RTO mandates also showed 18% higher patent application rates and 27% faster project completion times for collaborative initiatives. The study authors conclude that physical proximity is essential for innovation and that remote work arrangements stifle the spontaneous interactions that drive creative problem-solving. They recommend that organizations prioritize in-office requirements to maximize collaborative output and maintain competitive innovation capacity.
```
```

The analysis controlled for company size and industry sector. Notably, firms implementing RTO mandates were predominantly in the technology and financial services sectors, which may have different collaboration requirements than other industries.
```

**Gold Layer:**
```yaml
gold_label: overclaim
gold_confidence: high
gold_case_type: clear
gold_summary: |
  Severe selection bias (companies choosing RTO differ systematically from those maintaining remote work), reverse causation (struggling firms may impose RTO as intervention), conflation of correlation with causation, and inappropriate generalization from tech/finance to all industries.
gold_justification: |
  The study compares companies that chose RTO against their own prior performance, not against contemporaneous remote-first companies. Companies imposing RTO likely differ in unmeasured ways (management philosophy, financial stress, prior collaboration quality). The "control for company size and industry" is insufficient—RTO-imposing firms are self-selected and likely responding to specific performance issues (reverse causation). Patent applications and email volume are poor proxies for innovation quality; increased emails may indicate coordination friction rather than collaboration. The final caveat about tech/finance sectors is mentioned but ignored in the universal recommendation.
gold_evidence_span: "firms implementing RTO mandates were predominantly in the technology and financial services sectors"
gold_alternative_reading: |
  Companies imposing RTO may be those already struggling with collaboration or management trust issues. Email volume increases could reflect coordination friction requiring more communication overhead. Comparison lacks contemporaneous control group of remote-first firms.
```

Expected Findings:
```yaml
expected_findings:
  - finding: selection_bias
    severity: high
    confidence: high
  - finding: reverse_causation_possible
    severity: high
    confidence: medium
  - finding: proxy_measurement_issues
    severity: medium
    confidence: high
  - finding: inappropriate_generalization
    severity: high
    confidence: high
```
```

```

**Evidence Spans:**
```yaml
evidence_spans:
  - span: "companies that implemented mandatory in-office policies"
    why_it_matters: Self-selected sample; companies choosing RTO differ systematically from remote-first firms
    supports_finding: selection_bias
  - span: "23% more emails to colleagues"
    why_it_matters: Email volume is poor proxy for collaboration quality; may indicate increased coordination friction
    supports_finding: proxy_measurement_issues
  - span: "prioritize in-office requirements to maximize collaborative output"
    why_it_matters: Prescriptive recommendation based on correlational data without causal identification
    supports_finding: inappropriate_generalization
  - span: "firms implementing RTO mandates were predominantly in the technology and financial services sectors"
    why_it_matters: Explicit limitation ignored in universal recommendation
    supports_finding: selection_bias
```

Diagnostic Value:
```yaml
why_this_case_is_useful: |
  Tests recognition of selection bias in policy evaluation and the common fallacy of comparing self-selected adopters to non-adopters. Also tests ability to identify weak proxy measures (email volume as collaboration) and domain-inappropriate generalizations. Highly relevant to current workplace policy debates.
```

Case v0.2_005: Dopamine Habit Formation

Metadata:
```yaml
case_id: v0.2_005
title: "Dopamine Habit Formation"
domain: behavioral_neuroscience
category_bucket: mechanism_vs_evidence
difficulty: medium
source_type: synthetic_summary
expected_strength: weak
expected_main_failure: mechanistic_inflation
```
```

Input:
```yaml
input_text: |
  Neuroimaging research has elucidated the dopaminergic mechanisms underlying habit formation, providing a roadmap for behavioral change interventions. Using fMRI, researchers tracked dopamine release in the striatum of 24 participants during a sequential learning task over four weeks. The study identified distinct phasic dopamine signaling patterns that predicted successful habit automation, with increased dorsolateral striatum activity correlating with reduced cognitive effort during task execution.

  These findings establish the neurobiological basis of habit formation and demonstrate that dopaminergic pathways can be targeted to accelerate habit acquisition. The authors argue that behavioral interventions should leverage these dopamine dynamics to design more effective habit formation programs. They suggest that timing rewards to coincide with peak dopaminergic activity windows could significantly improve outcomes in addiction recovery, exercise adherence, and productivity systems.

  The study measured neural activity during a laboratory button-pressing task with monetary rewards. Participants completed an average of 12 sessions. No longitudinal behavioral outcomes were assessed beyond the laboratory setting.
```

Gold Layer:
```yaml
gold_label: mechanism_vs_evidence
gold_confidence: high
gold_case_type: clear
gold_summary: |
  Strong mechanistic evidence (neuroimaging showing dopamine-habit correlation) but massive inferential leap to clinical/behavioral efficacy without testing actual behavior change interventions or real-world durability.
gold_justification: |
  The study establishes correlation between neural activity and task learning in a highly artificial laboratory setting (button pressing for money). However, it makes unsupported leaps to clinical applications (addiction recovery) and complex real-world behaviors (exercise adherence) without evidence that the mechanism scales or translates. The phrase "establish the neurobiological basis" overstates what is essentially a correlation in 24 participants. The recommendation to "time rewards to coincide with peak dopaminergic activity windows" is speculative engineering based on observational neuroimaging with no intervention testing. Classic mechanism-evidence gap where biological plausibility substitutes for clinical proof.
gold_evidence_span: "behavioral interventions should leverage these dopamine dynamics"
gold_alternative_reading: |
  The dopamine patterns observed during artificial button-pressing tasks may not generalize to complex real-world behaviors. The correlation between striatal activity and task performance does not prove that manipulating dopamine timing would improve outcomes; could be epiphenomenal.
```

Expected Findings:
```yaml
expected_findings:
  - finding: mechanistic_inflation
    severity: high
    confidence: high
  - finding: generalization_from_lab_to_real_world
    severity: high
    confidence: high
  - finding: correlation_presented_as_intervention_target
    severity: high
    confidence: high
  - finding: small_sample_size
    severity: medium
    confidence: high
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "behavioral interventions should leverage these dopamine dynamics"
    why_it_matters: Prescriptive recommendation based on observational mechanism without intervention testing
    supports_finding: mechanistic_inflation
  - span: "could significantly improve outcomes in addiction recovery, exercise adherence"
    why_it_matters: Inappropriate generalization from button-pressing task to complex clinical outcomes
    supports_finding: generalization_from_lab_to_real_world
  - span: "24 participants"
    why_it_matters: Very small sample for mechanistic claims with broad generalization
    supports_finding: small_sample_size
  - span: "No longitudinal behavioral outcomes were assessed beyond the laboratory setting"
    why_it_matters: Explicit admission of evidence gap ignored in recommendations
    supports_finding: mechanistic_inflation
```

Diagnostic Value:
```yaml
why_this_case_is_useful: |
  Classic example of the mechanism-evidence gap common in neuroscience and behavioral psychology. Tests whether systems can distinguish between understanding biological mechanisms and having evidence for practical interventions. The "neuroplasticity," "dopamine," and "brain-based" rhetoric often obscures the lack of clinical efficacy data.
```

Case v0.2_006: Minimum Wage Automation

Metadata:
```yaml
case_id: v0.2_006
title: "Minimum Wage Automation"
domain: economics
category_bucket: correlation_vs_causation
difficulty: easy
source_type: synthetic_press_summary
expected_strength: weak
expected_main_failure: causal_overclaim
```

Input:
```yaml
input_text: |
  A comprehensive analysis of restaurant industry data reveals that minimum wage increases drive automation adoption and eliminate entry-level jobs. Examining county-level data from 2008 to 2019, the study found that jurisdictions raising minimum wages by 10% or more showed 34% higher rates of automated ordering kiosk installation and 28% greater adoption of kitchen automation technologies compared to counties without wage increases.

  Employment data showed that counties with aggressive minimum wage policies had 12% fewer fast-food workers per capita than control counties after five years. The study concludes that minimum wage mandates accelerate automation displacement of low-skill workers and reduce employment opportunities for entry-level positions. The authors recommend against further minimum wage increases, arguing that they harm the very workers they intend to help by forcing premature automation adoption.

  The analysis controlled for population density and median household income. Counties with minimum wage increases were primarily in coastal states with higher costs of living.
```

Gold Layer:
```yaml
gold_label: correlation_vs_causation
gold_confidence: high
gold_case_type: clear
gold_summary: |
  Observational correlation between minimum wage policies and automation confounded by secular technology trends, regional economic differences, and endogenous policy adoption. No causal identification strategy despite strong causal language.
gold_justification: |
  The study commits classic correlation-causation error. Counties raising minimum wages differ systematically from those that don't (coastal, higher cost of living, different labor markets). Automation adoption is occurring everywhere due to secular technology trends; attributing the difference entirely to minimum wage ignores confounding. The 12% employment difference over five years could reflect regional economic conditions rather than
```

causal effect. No difference-in-differences, instrumental variable, or regression discontinuity approach is mentioned. The policy recommendation against minimum wage increases is based on observational correlation presented as causal evidence of harm.
gold_evidence_span: "minimum wage increases drive automation adoption"
gold_alternative_reading: |
  Coastal counties with higher wages may have different technology adoption patterns independent of minimum wage policy. Automation is increasing everywhere; correlation with wage policy may reflect confounding by region or industry composition rather than causal effect.
```

**Expected Findings:**
```yaml
expected_findings:
  - finding: correlation_vs_causation
    severity: high
    confidence: high
  - finding: confounding_by_region
    severity: high
    confidence: high
  - finding: secular_trend_ignored
    severity: medium
    confidence: high
  - finding: inappropriate_policy_recommendation
    severity: high
    confidence: high
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "minimum wage increases drive automation adoption"
    why_it_matters: "Drive" implies causal mechanism based on observational correlation
    supports_finding: correlation_vs_causation
  - span: "Counties with minimum wage increases were primarily in coastal states"
    why_it_matters: Explicit admission of regional confounding ignored in causal claims
    supports_finding: confounding_by_region
  - span: "recommend against further minimum wage increases"
    why_it_matters: Policy prescription based on correlational evidence without causal identification
    supports_finding: inappropriate_policy_recommendation
```

Diagnostic Value:
```yaml
why_this_case_is_useful: |
  Clear example of causal overclaim in observational economics. Tests recognition that correlation in cross-sectional data does not imply causation, and that policy

recommendations require causal identification strategies. The "drive" verb is a strong causal claim unsupported by the study design.
```

Case v0.2_007: Universal Basic Income Pilots

Metadata:
```yaml
case_id: v0.2_007
title: "Universal Basic Income Pilots"
domain: public_policy
category_bucket: mixed
difficulty: hard
source_type: synthetic_summary
expected_strength: moderate_uncertainty
expected_main_failure: external_validity_uncertainty
```

Input:
```yaml
input_text: |
  Two-year randomized controlled trials of guaranteed income programs show complex effects on employment and wellbeing. The studies provided $1,000 monthly payments to 1,000 low-income participants in three US cities, with a control group of 1,000 participants receiving standard social services.

  Results showed a 5 percentage point reduction in full-time employment among recipients (42% vs 47% in control group), but this was offset by increased part-time work and education enrollment. Recipients reported significant improvements in financial security and stress reduction. However, the employment effects varied substantially by city: one site showed no employment reduction while another showed 9 percentage points decline.

  The studies were conducted during 2020-2022, a period of unusual labor market conditions. Participant dropout was higher than expected (18% vs projected 10%), and control group participants in two sites received additional pandemic-related benefits that blurred the treatment contrast. The authors note that results may not generalize to universal programs (these were targeted low-income populations) or to normal economic conditions.
```

Gold Layer:
```yaml
gold_label: mixed_evidence_boundary_case
gold_confidence: medium
gold_case_type: borderline
gold_summary: |
```
```

  Well-conducted RCT with genuine uncertainty about external validity, timing confounds, and heterogeneous effects. Reasonable analysts could disagree on whether this constitutes evidence for or against UBI policies.
gold_justification: |
  This is a legitimate borderline case. The RCT design is strong, but external validity concerns are severe: pandemic labor markets were anomalous, control group contamination occurred, attrition was high, and results were heterogeneous across sites. The 5 percentage point employment reduction is statistically significant but practically ambiguous (offset by education gains). A sympathetic analyst could argue this shows UBI enables education and doesn't cause mass workforce exit; a skeptical analyst could argue pandemic context obscured true displacement effects. Both readings are defensible. The authors' own uncertainty statements are appropriately cautious.
gold_evidence_span: "results may not generalize to universal programs"
gold_alternative_reading: |
  Sympathetic reading: Small employment reduction offset by human capital investment; financial security gains are real and meaningful. Critics: Pandemic context makes findings irrelevant to normal economies; targeted programs differ fundamentally from universal ones; 18% attrition threatens internal validity.
```

**Expected Findings:**
```yaml
expected_findings:
  - finding: external_validity_concerns
    severity: medium
    confidence: high
  - finding: temporal_confounding
    severity: medium
    confidence: high
  - finding: heterogeneous_effects
    severity: low
    confidence: high
  - finding: internal_validity_moderate
    severity: low
    confidence: medium
```

Evidence Spans:
```yaml
evidence_spans:
  - span: "conducted during 2020-2022, a period of unusual labor market conditions"
    why_it_matters: Temporal validity threat that creates genuine uncertainty about generalizability
    supports_finding: temporal_confounding
  - span: "control group participants in two sites received additional pandemic-related benefits"
    why_it_matters: Treatment contrast contamination complicates causal interpretation
    supports_finding: internal_validity_moderate
```
```

```
  - span: "varied substantially by city: one site showed no employment reduction while another showed 9 percentage points"
    why_it_matters: True effect heterogeneity or noise? Reasonable people disagree.
    supports_finding: heterogeneous_effects
  - span: "results may not generalize to universal programs"
    why_it_matters: Explicit boundary condition that makes this a legitimate uncertainty case
    supports_finding: external_validity_concerns
```

Diagnostic Value:
```yaml
why_this_case_is_useful: |
  Tests ability to hold epistemic uncertainty without forced over-certainty. Distinguishes between "this is bad science" (Cases 3-6) and "this is good science with genuinely ambiguous implications." Critical for evaluating whether forensic systems can grade on a curve—recognizing when uncertainty is appropriate versus when certainty is possible.
```

---

4. Coverage Analysis

Improvements over v0.1 Pilot

v0.1 Weakness	v0.2 Improvement
Weak overclaim examples (vitamin D ambiguity)	Cases 3 & 4 feature clear, sharp overclaims with unambiguous selection bias and rhetorical inflation
Heavy health/supplement focus	Domain expansion to education, environmental science, technology, workplace policy, economics, and public policy
Missing borderline cases	Case 7 provides explicit uncertainty boundary with defensible alternative readings
Sparse gold justifications	Every case includes explicit justification for labels, not just label assignment
Messy source artifacts in input text	All input_text is clean prose with no citations, URLs, or contentReference artifacts
Limited mechanism-evidence distinction	Case 5 specifically targets the neuroscience mechanism-to-intervention gap

Failure Mode Coverage

Well-covered:
- **Survivorship bias** (Case 3): Technology adoption studies
- **Selection bias/Reverse causation** (Case 4): Policy evaluation confounds
- **Mechanistic inflation** (Case 5): Neuroscience-to-clinical leaps
- **Correlation-causation** (Case 6): Observational econometrics
- **External validity uncertainty** (Case 7): Pilot program generalization

Gaps for v0.3:
- **Publication bias/meta-analysis issues**: No cases examining file-drawer effects or selective outcome reporting
- **P-hacking/fishing expeditions**: No cases with multiple comparison problems or questionable research practices
- **Replication failures**: No cases comparing original dramatic findings to failed replications
- **Industry funding bias**: No cases examining conflicts of interest in pharmaceutical or nutrition research
- **Statistical power/Type M errors**: No cases highlighting underpowered studies with inflated effect sizes

Diagnostic Distribution

Diagnostic Target	Cases
Recognizing appropriate caution	1, 2
Spotting rhetorical inflation	3, 4, 6
Mechanism vs. evidence separation	5
Holding epistemic uncertainty	7

This pack provides a **tight, high-signal expansion** that maintains the forensic focus of Agent Arena while broadening domain coverage and sharpening the overclaim examples. The inclusion of Case 7 as an explicit borderline example ensures the dataset tests appropriate uncertainty calibration, not just error detection.